

1.0 COURSEWORK TITLE

COURSE RECOVERY SYSTEM (CRS) FOR EDUCATIONAL INSTITUTIONS

2.0 THE COURSEWORK OVERVIEW

CRS is a software solution that is designed to help educational institutions to manage how the students recover failed or incomplete courses. It ensures that the students can resume on track academically without delaying graduation. The key purpose of the solution **are** to allow students to enrol in recovery process to improve academic profiles and also allow course administrators and instructors to plan, track and evaluate course recovery plan. To streamline this academic study workflow, a Java-based application must be developed using Object-Oriented Programming (OOP) principles. The system should offer the functional requirements as stated in the section 5.0. In addition, a program documentation is needed covering solution model design and the implementation details that utilises the Object-oriented programming concepts.

3.0 OBJECTIVES OF THIS COURSEWORK

Develop the practical ability to describe, justify, and implement an Object-oriented system.

4.0 TYPE

Group Assignment (4 or 5 students)

5.0 COURSEWORK DESCRIPTION

The developed system should be GUI-driven with options. You and your development team as Object-oriented programmers need to identify the relationship among the object entities followed by designing and developing the necessary classes with appropriate relationships to fulfil the requirements of the expected systems.

Target Users: Academic officer and Course Administrator

Functional requirements:

1) User Management

This is a process of controlling and maintaining users' accounts and accessibility and interaction with the system. The following features are commonly included, but not limited to.

- User accounts (Add/Update/Deactivate)
- Authentication and Authorization
 - Login/Logout, log the timestamp of both in binary form
 - Role-based permission enforcement or access control
- Password and Credential Management (Reset/Recover)

2) Course Recovery Plan

This is a special strategy process that allows students to improve weak performance, through recovering the failed modules with specified timeline and deadlines. The following features are commonly included, but not limited to.

- List out all failed components of the course i.e., assignment or exam for the student concerned
- Recommendation entry for the course recovery (Add/Update/Remove)
- Set a clear milestone for course recovery with the action plan (Add/Update/Remove)
 - Example,

COURSE	STUDY WEEK	TASK
Object Oriented Programming	Week 1 – 2	Review all lecture topic
	Week 3	Meeting with module lecturer
	Week 4	Take recovery exam

- Monitor/ Track the Recovery progress inclusive the grading entry

3) Eligibility Check and Enrolment

This is a process of checking whether the students meet the required criteria to join the next level of study. The following features are commonly included, but not limited to.

- List out all students who are *not* eligible to progress into next level of study. The two eligibility criteria of progression as follows:

- At least CGPA 2.0, the Credit Hour System as such

$$\text{CGPA} = (\text{Total Grade Points of all courses taken}) / (\text{Total Credit Hours of all courses taken})$$

Computation sample:

- Course 1: A (4.0) × 3 credits = 12
- Course 2: B (3.0) × 4 credits = 12
- Course 3: C (2.0) × 2 credits = 4
- Total = 28 / 9 = 3.11 CGPA

- Not more than three failed courses to be allowed to progress into next level of study

- To allow registration once the eligibility is confirmed.

4) Academic Performance Reporting

This is the formal document that provides a summary of a student's progress, achievements, and the recommendations of improvement over a specific period. The following features are commonly included, but not limited to.

- Generate an academic performance report by semester and year of study (create/exportToPDF)
- Example:
Student Name: Alex Tan
Student ID: 2025A1234
Program: Bachelor of Computer Science
Semester 1

Course Code	Course Title	Credit Hours	Grade	Grade Point (4.0 Scale)
CS201	Data Structures	3	A	4.0
CS205	Database Systems	3	B+	3.3
CS210	Software Engineering I	3	B	3.0
MA202	Discrete Mathematics	4	C+	2.3
EN201	Academic Writing	2	A-	3.7

Cumulative GPA (CGPA): 3.25

5) Email Notifications

This is an automated messages send by a system to the users i.e., student's email address. It helps maintain communication message as alerts, reminders, confirmations and etc between the system and its users to improve accountability. The following are the applicable areas for this service, but not limited to.

- User account management
- Password & recovery management
- Course recovery with a clear action plan and milestone
- Academic performance report

Technical Consideration:

1) Solution design practice:

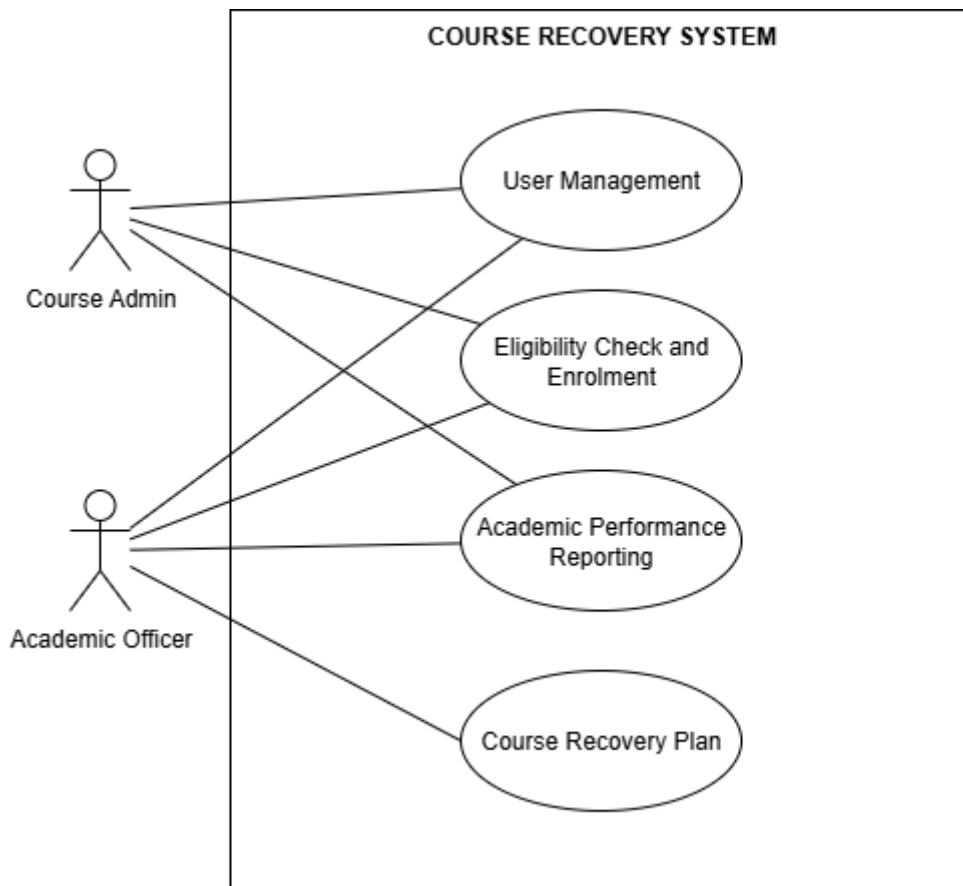
This is a collection of Object-oriented principles, modelling, best practices and documentation standards used to design and develop a software. The following are the development practices for the solution design.

- Package
- Object & Class
- Inheritance hierarchy
- Encapsulation
- Polymorphism
- Abstraction
- Modularity

2) Third party APIs:

- Java Mail API for the email notification services, <https://www.oracle.com/java/technologies/javamail-api.html>
- iText PDF, <https://itextpdf.com/resources/downloads>

The following are the additional descriptions for better understanding of the academic study system.

Overview Use Case Diagram:**Course Retrieval Policy:**

- Each course has three attempt opportunities to pass the course **only**.
- The failed component is required **only** to pass by resubmission or resit in 2nd attempt.
- If the students cannot pass the course in the 1st and 2nd attempts, the 3rd attempt requires the students to refer to all assessment components.

6.0 **GENERAL REQUIREMENTS**

- The program submitted should compile and be executed without errors
- Validation should be done for each entry from the users in order to avoid logical errors.
- The implementation code must highlight the use of Object-oriented programming concepts as required by the solution.
- Use text or binary files for data access and manipulation only.

7.0 **DELIVERABLES:**

- Java Program Documentation in softcopy form
- Working GUI-driven software prototype
- Software Demonstration or Presentation
- Submission deadline: **TO BE SPECIFIED IN MOODLE**

7.1 ***DOCUMENTS: COURSEWORK REPORT***

- ✓ As part of the assessment, you must submit the project report in softcopy form, which should have the following format:

A) Cover Page:

All reports must be prepared with a *front cover*. A protective transparent plastic sheet can be placed in front of the report to protect the front cover. The front cover should be presented with the following details:

- ↗ Module
- ↗ Coursework Title
- ↗ Intake
- ↗ Student name and id
- ↗ Date Assigned (the date the report was handed out).
- ↗ Date Completed (the date the report is due to be handed in).

B) Contents:

- ↗ Description and justification of Object-oriented programming concepts incorporated into the solution
- ↗ Design solution
- ↗ Screenshots of output of the program with appropriate explanations

C) Limitation and Conclusion

D) References

8.0 ASSIGNMENT ASSESSMENT CRITERIA

The assignment assessment consists of three components: Program Documentation (15 marks), Program implementation (40 marks). Details of the division for each component are as follows:

CRITERIA	MARK ALLOCATION
<i>Program Documentation:</i>	
Detail Use Case Diagram with description	5 marks
Class Diagram	5 marks
Report Content and Presentation or Demonstration	5 marks
SUBTOTAL	15 marks
<i>Program Implementation:</i>	
User Management	5 marks
Eligibility Check and Enrolment	5 marks
Academic Performance Reporting	5 marks
Course Recovery Plan	10 marks
Object Oriented Implementation	15 marks
SUBTOTAL	40 marks
TOTAL	55 marks

9.0 DEVELOPMENT TOOLS

The program written for this assignment should be written applying Object Oriented Programming using JAVA language only.

10.0 ACADEMIC INTEGRITY

- You are expected to maintain the utmost level of academic integrity during the duration of the course.
- Plagiarism is a serious offence and will be dealt with according to APU University regulations on plagiarism.